Duality in Action

Stationary Distributions, Piecewise-Linear Optimization, and Robust Linear Programs

A study guide for Homework 4

Optimization — companion notes | September 27, 2026

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Key idea. Homework 1 supplied the geometry: convex sets, convex combinations, epigraphs, and separating hyperplanes. Homework 3 supplied certificates: duality, complementary slackness, and theorems of the alternative. Homework 4 uses these tools to prove existence, minimize a maximum, and replace infinitely many requirements by a finite LP.

How to use this guide. Read Sections 1–2, work the problems with the separate solutions closed, and then use the relevant application section to check your reasoning. Finish with the self-tests and the one-page reference sheet. Optional Problem 5 is included as an extension.

Source: supplied `hw04.pdf`, pp. 1–3. These companion notes contain original explanations and examples, not an official solution key. Numerical checks illustrate the theory; they do not replace proofs.

1 The duality toolkit: from geometry to certificates

Throughout, vector inequalities are componentwise. A variable without a stated sign restriction is *free*. Let $A \in \mathbb{R}^{m \times n}$.

A feasible solution gives one bound; a dual solution gives the other. For the pair

$$\min\{c^\mathsf{T} x : Ax \geq b,\ x \geq 0\}, \qquad \max\{b^\mathsf{T} y : A^\mathsf{T} y \leq c,\ y \geq 0\},$$

feasibility gives $b^\mathsf{T} y \leq y^\mathsf{T} Ax \leq c^\mathsf{T} x$. This is weak duality. If these values agree, both solutions are optimal. For LPs, a feasible problem with a finite optimal value attains that value, and its dual has an optimizer with the same value: strong duality.

Fact	**What it lets you do**
Weak duality	Certify a bound or optimality of a feasible pair.
Strong duality	Prove that some exact certificate exists.
Complementary slackness	Identify which constraints can carry nonzero weights.
Theorems of the alternative	Turn failure of feasibility into a separating certificate.

Rebuild signs from the Lagrangian. For a minimization problem with inequalities $g_i(z) \leq 0$, use $L(z, p) = f(z) + \sum_i p_i g_i(z)$ with $p_i \geq 0$. For feasible z, $L(z, p) \leq f(z)$, so $\inf_z L$ is a lower bound. A free variable with a nonzero linear coefficient makes that infimum $-\infty$; its coefficient must vanish.

Three pairs used repeatedly in this assignment:

$$\begin{aligned}
\max\{q^\mathsf{T} z : Bz = h,\ z \geq 0\} &\longleftrightarrow \min\{h^\mathsf{T} y : B^\mathsf{T} y \geq q,\ y \text{ free}\},\\
\max\{q^\mathsf{T} z : Bz \leq h,\ z \geq 0\} &\longleftrightarrow \min\{h^\mathsf{T} y : B^\mathsf{T} y \geq q,\ y \geq 0\},\\
\min\{t : Ax - t\mathbf{1} \leq b\} &\longleftrightarrow \max\{-b^\mathsf{T} p : A^\mathsf{T} p = 0,\ \mathbf{1}^\mathsf{T} p = 1,\ p \geq 0\}.
\end{aligned}$$

Both x and t are free in the last pair.

Common pitfall. Dual infeasibility alone does not imply primal unboundedness: the primal might itself be infeasible. In Problem 1, explicitly checking primal feasibility at $\lambda = 0$ is what makes the argument work.

The recurring pattern. Write a family of inequalities, combine them with nonnegative weights, and make the unwanted variable disappear. The remaining expression is a certificate that can be checked directly.

2 Stochastic matrices: existence from duality (Problem 1)

A row-stochastic matrix satisfies $p_{ij} \geq 0$ and $\sum_j p_{ij} = 1$. Think of p_{ij} as the probability of moving from state i to state j. If the current distribution is a *column* π, the next distribution is $P^\mathsf{T}\pi$.

An invariant distribution is a fixed point of this update:

$$P^\mathsf{T}\pi = \pi, \qquad \pi \geq 0, \qquad \mathbf{1}^\mathsf{T}\pi = 1.$$

First find a nonzero nonnegative invariant vector, then normalize it. This removes the normalization constraint while proving existence.

Recipe. Turn existence into boundedness, and then contradict the dual.

1. Form $\max \mathbf{1}^\mathsf{T}\lambda$ subject to $(P^\mathsf{T} - I)\lambda = 0$, $\lambda \geq 0$. The zero vector is feasible.
2. If no nonzero feasible vector exists, the optimal value is 0.
3. Its dual is $\min 0$ subject to $(P - I)y \geq \mathbf{1}$, with y free. Strong duality would require a feasible dual solution.
4. Choose k with $y_k = \max_j y_j$. Then $(Py)_k = \sum_j p_{kj} y_j \leq y_k$, so the kth dual inequality is impossible.

Therefore a nonzero $\lambda \geq 0$ exists, and $\pi = \lambda/(\mathbf{1}^\mathsf{T}\lambda)$ is an invariant probability distribution.

Why the maximum-coordinate trick works. Each row of P forms a convex combination of the entries of y. A weighted average cannot exceed the largest entry. Consequently, P cannot increase *every* coordinate of any vector by at least 1.

Key idea. This connects Homework 1's convex-combination inequality to Homework 3's duality. Geometry rules out a dual certificate; strong duality then forces the invariant vector to exist.

Equivalent theorem-of-the-alternative view. For a matrix M, Gordan's alternative says exactly one of the following is possible:

$$\exists y: \ My > 0, \qquad \text{or} \qquad \exists \lambda \geq 0, \ \lambda \neq 0: \ M^\mathsf{T}\lambda = 0.$$

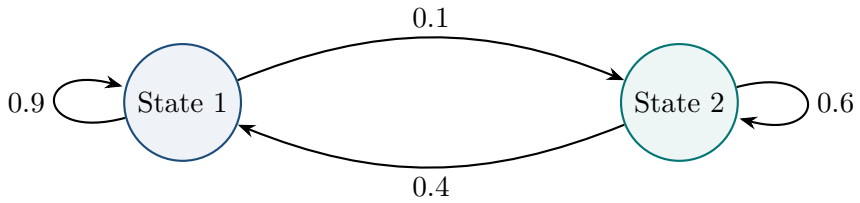
With $M = P - I$, the maximum-coordinate argument rules out the first case. The LP proof above supplies the same conclusion using ordinary LP duality.

Common pitfall. $P\mathbf{1} = \mathbf{1}$ proves the existence of a *right* eigenvector. It does not by itself provide the nonnegative *left* eigenvector needed for $P^\mathsf{T}\pi = \pi$. Keep the transpose and probability convention visible.

3 Markov-chain intuition and what existence does not prove

Consider the two-state chain

$$P = \begin{pmatrix} 0.9 & 0.1 \\ 0.4 & 0.6 \end{pmatrix}.$$



At stationarity, net probability flow between the states balances: $0.1\pi_1 = 0.4\pi_2$. Together with $\pi_1 + \pi_2 = 1$, this gives

$$\pi = (0.8, 0.2)^\mathsf{T}, \qquad P^\mathsf{T}\pi = (0.9 \cdot 0.8 + 0.4 \cdot 0.2,\ 0.1 \cdot 0.8 + 0.6 \cdot 0.2)^\mathsf{T} = \pi.$$

Pairwise flow balance is useful here because there are only two states; it is not a required property of every stationary distribution in a larger chain. Stationarity requires total inflow to equal total outflow at each state.

Connection to convex feasibility. The probability simplex is $\Delta_n = \{\pi \geq 0 : \mathbf{1}^\mathsf{T}\pi = 1\}$. The stationary distributions form

$$\Delta_n \cap \ker(P^\mathsf{T} - I),$$

a convex polytope. If π and ρ are stationary, every mixture $\theta\pi + (1-\theta)\rho$ is stationary too. The theorem says this intersection is nonempty.

Existence, uniqueness, and convergence are different claims.

- **Existence always holds** for a finite row-stochastic matrix.
- **Uniqueness can fail.** If $P = I$, every probability vector is stationary.
- **Convergence can fail even with uniqueness.** If $P = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$, the unique stationary distribution is $(1/2, 1/2)$, but a chain started at $(1, 0)$ alternates between $(1, 0)$ and $(0, 1)$.

As a later connection, finite irreducible chains have a unique stationary distribution; adding aperiodicity gives convergence from every initial distribution. These are stronger results than the homework asks you to prove.

Connection to reinforcement learning. Under a fixed stationary policy, a finite-state environment induces a transition matrix. An invariant distribution describes a probability flow that remains unchanged under that policy. Whether the distribution from an arbitrary initial state converges to it needs the additional assumptions above.

4 Minimizing a maximum: the epigraph LP (Problem 2)

Let $f(x) = \max_i(a_i^\mathsf{T} x - b_i)$, with rows a_i^T collected in A. Each affine piece is convex; a pointwise maximum of convex functions is convex. It may be nondifferentiable where several pieces meet.

The epigraph is a polyhedron. A scalar t is above the maximum exactly when it is above every piece:

$$t \geq f(x) \iff a_i^\mathsf{T} x - b_i \leq t \quad \forall i.$$

Hence

$$\min_x f(x) = \min_{x,t}\{t : Ax - t\mathbf{1} \leq b\}, \qquad x, t \text{ free.}$$

This is Homework 1's epigraph idea used as a modeling tool.

Derive the dual, rather than memorize it. For $p \geq 0$,

$$L(x,t,p) = t + p^\mathsf{T}(Ax - t\mathbf{1} - b) = (A^\mathsf{T}p)^\mathsf{T}x + (1 - \mathbf{1}^\mathsf{T}p)t - b^\mathsf{T}p.$$

To avoid $\inf_{x,t} L = -\infty$, both free-variable coefficients must vanish:

$$\boxed{A^\mathsf{T}p = 0, \qquad \mathbf{1}^\mathsf{T}p = 1, \qquad p \geq 0.}$$

The remaining constant is $-b^\mathsf{T}p$. Thus these weights give the dual $\max -b^\mathsf{T}p$ under the displayed constraints.

What each constraint means:

- $p \geq 0$: combine valid inequalities without reversing them.
- $\mathbf{1}^\mathsf{T}p = 1$: a convex combination cannot exceed the largest piece.
- $A^\mathsf{T}p = 0$: balance the slopes so the lower bound does not depend on x.

Indeed, for every feasible p and every x,

$$f(x) \geq \sum_i p_i(a_i^\mathsf{T} x - b_i) = -b^\mathsf{T}p.$$

The epigraph LP is feasible. Under the assumed finite value, LP attainment and strong duality show that the best such lower bound equals the minimum.

Common pitfall. Do not add $t \geq 0$. The maximum of affine functions can have a negative minimum. Likewise, adding $x \geq 0$ changes the homework and changes the dual condition $A^\mathsf{T}p = 0$ to an inequality.

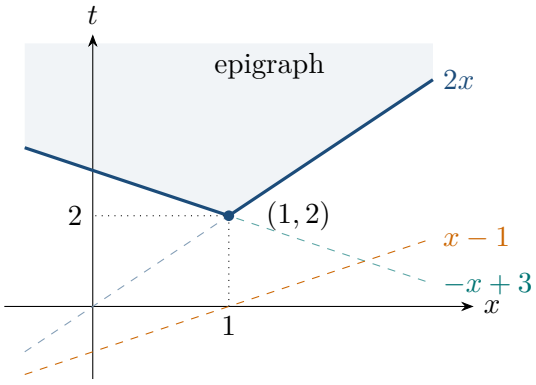
A useful modeling extension. Chebyshev fitting, $\min_x \|Cx - d\|_\infty$, is also an LP: minimize t subject to $-t\mathbf{1} \leq Cx - d \leq t\mathbf{1}$. It is the maximum of the affine residuals and their negatives, so the same epigraph method applies.

5 Active slopes, subgradients, and a worked example

Take

$$f(x) = \max\{2x,\ -x + 3,\ x - 1\}.$$

The first two pieces meet at $x^* = 1$, with $f(1) = 2$; the third has value 0.



Choose $p = (1/3, 2/3, 0)$. Its entries sum to 1, and $2p_1 - p_2 + p_3 = 0$. Since $b = (0, -3, 1)$, the dual lower bound is $-b^\mathsf{T} p = 2$. The candidate attains this bound, proving optimality.

Why only active slopes matter. At an optimal pair, complementary slackness says

$$p_i^* [t^* - (a_i^\mathsf{T} x^* - b_i)] = 0.$$

An inactive piece has positive slack and therefore zero weight. If $I(x) = \{i : a_i^\mathsf{T} x - b_i = f(x)\}$, optimality is equivalent to

$$\boxed{0 \in \operatorname{conv}\{a_i : i \in I(x)\}.}$$

In the example, $0 = (1/3) \cdot 2 + (2/3) \cdot (-1)$.

Connection to subgradients. A subgradient g at x satisfies $f(z) \geq f(x) + g^\mathsf{T}(z - x)$ for all z. For a finite maximum of affine functions, $\partial f(x) = \operatorname{conv}\{a_i : i \in I(x)\}$. Thus $0 \in \partial f(x)$ is the nonsmooth analogue of $\nabla f(x) = 0$. This previews first-order optimization methods.

Connection to minimax. Since a maximum equals the maximum over probability weights, $f(x) = \max_{p \in \Delta_m} p^\mathsf{T}(Ax - b)$. Strong LP duality gives $\inf_x \max_{p \in \Delta_m} p^\mathsf{T}(Ax - b) = \max_{p \in \Delta_m} \inf_x p^\mathsf{T}(Ax - b)$ when the value is finite. The opponent's optimal mixture cancels all dependence on x.

6 Symmetry and certificates of optimality (Problem 3)

Consider

$$\min\{c^\mathsf{T} x : Ax \geq c,\ x \geq 0\}, \qquad A = A^\mathsf{T}.$$

If $Ax^* = c$ and $x^* \geq 0$, the same vector works on both sides of the primal–dual pair:

$$\text{dual:} \quad \max\{c^\mathsf{T} y : A^\mathsf{T} y \leq c,\ y \geq 0\}, \qquad y^* = x^*.$$

It is feasible because $A^\mathsf{T} x^* = Ax^* = c$, and the objectives agree. Weak duality proves optimality immediately.

Read the proof as a weighted sum of constraints. For any feasible x,

$$\underbrace{c^\mathsf{T} x}_{\text{candidate cost}} = (Ax^*)^\mathsf{T} x = (x^*)^\mathsf{T} Ax \geq (x^*)^\mathsf{T} c = \underbrace{c^\mathsf{T} x^*}_{\text{certified lower bound}}.$$

The nonnegative entries of x^* are precisely the weights applied to $Ax \geq c$. Symmetry makes their weighted left side equal the objective.

Key idea. Symmetry is used to move A across the inner product. Positive semidefiniteness is not needed: the objective and constraints are linear, so the problem is an LP regardless of the eigenvalues of A.

An indefinite example. Let $A = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ and $c = (1, 1)$. Its eigenvalues are 1 and -1, but $x^* = (1, 1)$ is feasible with $Ax^* = c$. The constraints are $x_2 \geq 1$, $x_1 \geq 1$, so the minimum of $x_1 + x_2$ is 2.

Why symmetry matters. Take $A = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$, $c = (2, 1)$, and $x^* = (1, 1)$. Although $Ax^* = c$ and $x^* \geq 0$, its cost is 3. The feasible point $(0, 2)$ costs 2. The proposed certificate fails because $A^\mathsf{T} x^* = (1, 2) \not\leq (2, 1)$.

Connection to complementary slackness and KKT. For a general feasible pair (x, y) in this LP, the gap decomposes as

$$c^\mathsf{T} x - c^\mathsf{T} y = x^\mathsf{T}(c - A^\mathsf{T} y) + y^\mathsf{T}(Ax - c) \geq 0.$$

At the special pair $x = y = x^*$, both slack vectors vanish. This is the linear-programming version of the KKT optimality conditions: feasibility, dual feasibility, stationarity, and complementary slackness.

Common pitfall. Tight primal constraints alone do not prove optimality. You must also produce a valid bound, such as a dual-feasible vector with matching cost. Nor does this argument claim the optimizer is unique.

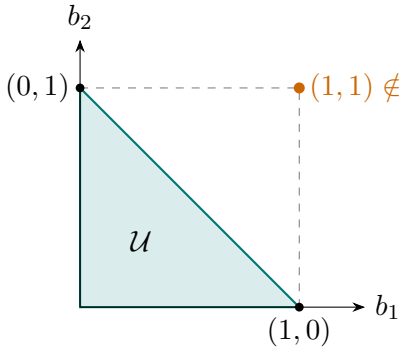
7 Robust right-hand sides and the order of quantifiers (Problem 4)

A robust constraint must hold for *every* allowed uncertainty value. The decision x is chosen once, before the uncertainty is revealed. For a nonempty uncertainty set $\mathcal{U}$,

$$Ax \geq b \ \forall b \in \mathcal{U} \iff (Ax)_i \geq \sup_{b \in \mathcal{U}} b_i \quad \forall i.$$

For $Ax \leq b$, use the coordinatewise *infimum* instead.

In Problem 4, the displayed set is $\mathcal{U} = \{b \geq 0 : \mathbf{1}^\mathsf{T} b \leq 1\}$; the extra $b_i \leq 1$ bounds are redundant. It is a simplex, despite being called a unit box. Each $e_i \in \mathcal{U}$, so each coordinate can individually reach 1. The exact counterpart is therefore $Ax \geq \mathbf{1}$.



For two rows, scenario $(1,0)$ forces the first threshold to 1, while $(0,1)$ forces the second threshold to 1. The two worst cases need not occur in the same scenario. Universal quantifiers commute: $\forall b \, \forall i$ is the same as $\forall i \, \forall b$.

Common pitfall. Do not replace the vector constraint by $\sum_i (Ax)_i \geq \max_{b \in \mathcal{U}} \sum_i b_i = 1$. That is weaker: $(Ax) = (1,0)$ passes the sum test but fails against $b = (0,1)$. Row-by-row guarantees are essential.

Why the uncertainty budget disappears here. With $\mathcal{U}_\Gamma = \{0 \leq b \leq \mathbf{1} : \mathbf{1}^\mathsf{T} b \leq \Gamma\}$, $\Gamma \geq 0$, the answer is $Ax \geq \min\{1, \Gamma\}\mathbf{1}$. Once $\Gamma \geq 1$, each coordinate can already hit its individual maximum. In a *single* constraint containing a sum of uncertain terms, the budget can still matter substantially.

8 Uncertain coefficients: support functions and dual LPs

For coefficient uncertainty, the uncertain vector multiplies the decision:

$$a^\mathsf{T} x \le \beta \quad \forall a \in \mathcal{U} \quad \Longleftrightarrow \quad \sigma_{\mathcal{U}}(x) := \sup_{a \in \mathcal{U}} a^\mathsf{T} x \le \beta.$$

$\sigma_{\mathcal{U}}$ is the *support function*. It is convex because it is a pointwise supremum of linear functions of x; it may equal $+\infty$ for an unbounded set. This directly connects robustness to Homework 1's convex-function calculus.

Equality constraints and nonnegative uncertain coefficients. For nonempty $\mathcal{U}_2 = \{a : D_2 a = d_2,\ a \ge 0\}$, dualize the adversary's maximization with x held fixed:

$$\sup_{a \ge 0,\ D_2 a = d_2} x^\mathsf{T} a \quad \longleftrightarrow \quad \inf_{y_2 \text{ free},\ D_2^\mathsf{T} y_2 \ge x} d_2^\mathsf{T} y_2.$$

The exact robust counterpart is the *existence* of a certificate:

$$\boxed{\exists y_2 \text{ free}: \quad D_2^\mathsf{T} y_2 \ge x, \quad d_2^\mathsf{T} y_2 \le \beta.}$$

Weak duality proves sufficiency: for every allowed a, $x^\mathsf{T} a \le y_2^\mathsf{T} D_2 a = y_2^\mathsf{T} d_2 \le \beta$. If robustness holds, the nonempty inner LP has finite value, so strong duality supplies a certificate and proves necessity.

Inequality constraints and nonpositive uncertain coefficients. For $\mathcal{U}_3 = \{a : D_3 a \ge d_3,\ a \le 0\}$, first use $z = -a$:

$$\sup_{z \ge 0,\ D_3 z \le -d_3} (-x)^\mathsf{T} z \quad \longleftrightarrow \quad \inf_{\mu_3 \ge 0,\ D_3^\mathsf{T} \mu_3 \ge -x} (-d_3)^\mathsf{T} \mu_3.$$

Thus, for nonempty $\mathcal{U}_3$,

$$\boxed{\exists \mu_3 \ge 0: \quad D_3^\mathsf{T} \mu_3 \ge -x, \quad -d_3^\mathsf{T} \mu_3 \le \beta.}$$

Recipe. Convert a robust coefficient constraint to finite constraints.

1. Freeze x and maximize the uncertain left-hand side over its set.
2. Normalize signs, such as $z = -a \ge 0$, before forming the dual.
3. Introduce the dual variables as extra decision variables.
4. Require dual feasibility and a dual objective no larger than the RHS.
5. Justify exactness using nonemptiness, bounded value, and strong duality.

The new variables are certificates of robustness, not extra uncertain quantities. No additional sign restriction on x is implied.

9 The optional problem, edge cases, and broader connections

Resolve the handout's notation. For $A \in \mathbb{R}^{m\times n}$ and $x \in \mathbb{R}^n$, use $Ax \le b$ for the first block of Problem 5. The printed $A^\mathsf{T}x$ needs $A \in \mathbb{R}^{n\times m}$ instead. The final set is $\mathcal{U}_3$, correcting the repeated $\mathcal{U}_2$ label.

For a nonempty polytope $\mathcal{U}_1$, precompute $\underline{b}_i = \min_{b\in\mathcal{U}_1} b_i$. With all sets nonempty, the finite LP is

$$\begin{aligned} \min_{x,y_2,\mu_3} \quad & c^\mathsf{T}x \\ \text{s.t.} \quad & Ax \le \underline{b}, \\ & D_2^\mathsf{T}y_2 \ge x, \qquad d_2^\mathsf{T}y_2 \le b_2, \\ & D_3^\mathsf{T}\mu_3 \ge -x, \qquad -d_3^\mathsf{T}\mu_3 \le b_3, \\ & \mu_3 \ge 0, \qquad x, y_2 \text{ free}. \end{aligned}$$

If D_j has r_j rows, its certificate vector has r_j entries. The problem uses $n + r_2 + r_3$ variables. The coordinatewise minima need not all occur at the same $b \in \mathcal{U}_1$.

Two small sets that make the formulas tangible.

1. If $\mathcal{U}_2 = \{a \ge 0 : a_1 + a_2 = 1\}$, its support function is $\max(x_1, x_2)$. The dual certificate is a scalar y satisfying $y \ge x_1, x_2$ and $y \le \beta$.
2. If $\mathcal{U}_3 = \{a \le 0 : a_1 + a_2 \ge -1\}$, its vertices are $0, -e_1, -e_2$. Its support is $\max(0, -x_1, -x_2)$. The robust constraint is $\beta \ge 0$ and $x_1, x_2 \ge -\beta$.

Empty sets. A universal statement over an empty set is true. Check emptiness by LP feasibility, then omit the associated constraint block if its uncertainty set is empty.

Unbounded sets. With $\mathcal{U} = \{a \ge 0 : a_1 = 1\}$, $\sigma_\mathcal{U}(x) = x_1$ if $x_2 \le 0$ and $+\infty$ if $x_2 > 0$. The dual automatically enforces $x_2 \le 0$. An unbounded uncertainty set can therefore still permit robust decisions.

Beyond polytopes: uncertainty geometry chooses the model. For $a = a_0 + u$ with $\|u\|_2 \le \rho$, Cauchy–Schwarz gives $\sup a^\mathsf{T}x = a_0^\mathsf{T}x + \rho\|x\|_2$. Robustness becomes a second-order cone constraint. For $\|u\|_\infty \le \rho$, the answer is $a_0^\mathsf{T}x + \rho\|x\|_1$, which can be modeled by an LP with absolute-value auxiliaries. These are consequences of the uncertainty set's support function.

10 Self-tests: work these before opening the answers

Self-test 1. *Let* $P = \left(\begin{smallmatrix} 3/4 & 1/4 \\ 1/2 & 1/2 \end{smallmatrix}\right)$. *Find its stationary distribution and verify it by multiplication.*

Self-test 2. *For* $P = I$, *how many stationary distributions exist? For the two-state swap matrix, does a unique stationary distribution imply convergence from* $(1, 0)$*? Explain the distinction in one sentence.*

Self-test 3. *Write the epigraph LP and its dual for* $\min_x \max\{x+1, -2x+4\}$. *Find a primal optimum and dual certificate.*

Self-test 4. *For* $f(x) = \max\{x, 2x\}$, *both pieces are active at* $x = 0$. *Is* 0 *a minimizer? What does the convex hull of the active slopes say?*

Self-test 5. *For a finite maximum of affine functions on all of* $\mathbb{R}^n$, *show that the value is finite if and only if* $0 \in \operatorname{conv}\{a_1, \ldots, a_m\}$. *Hint: use a feasible dual mixture in one direction and strict separation from a compact convex hull in the other.*

Self-test 6. *Let* $A = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$, $c = (2, 3)$. *Find* $x^* \geq 0$ *with* $Ax^* = c$ *and certify the minimum of* $c^\mathsf{T} x$ *over* $Ax \geq c$, $x \geq 0$.

Self-test 7. *Let* $\mathcal{U} = \{b \geq 0 : b_1 + b_2 \leq 1\}$. *Give the exact counterparts of (i)* $Ax \geq b$ *for all* $b \in \mathcal{U}$ *and (ii)* $Ax \leq b$ *for all* $b \in \mathcal{U}$. *Explain why the two answers differ.*

Self-test 8. *If* $\mathcal{U} = \{a \geq 0 : a_1 + a_2 = 1\}$, *eliminate uncertainty from* $a^\mathsf{T} x \leq 5$. *Derive the same result through a scalar dual variable.*

Self-test 9. *If* $\mathcal{U} = \{a \leq 0 : a_1 + a_2 \geq -1\}$, *eliminate uncertainty from* $a^\mathsf{T} x \leq \beta$. *Can any* x *work if* $\beta = -1$*?*

Self-test 10. *If* $\mathcal{U} = \{a \geq 0 : a_1 = 1\}$, *when does* $a^\mathsf{T} x \leq 3$ *hold for all* $a \in \mathcal{U}$*? What changes if the uncertainty set is empty?*

Self-test 11. *Suppose* $a = a_0 + u$, $\|u\|_2 \leq 2$. *Write the exact counterpart of* $a^\mathsf{T} x \leq 7$. *Which Homework 1 inequality proves it?*

11 Self-test answers and proof sketches

1. Flow balance gives $\frac{1}{4}\pi_1 = \frac{1}{2}\pi_2$, so $\pi = (2/3, 1/3)$. Directly, $P^\mathsf{T}\pi = (\frac{3}{4}\frac{2}{3}+\frac{1}{2}\frac{1}{3}, \frac{1}{4}\frac{2}{3}+\frac{1}{2}\frac{1}{3}) = (2/3, 1/3)$.

2. Every probability vector is stationary for I. The swap matrix has the unique stationary vector $(1/2, 1/2)$, yet the stated initial distribution alternates. A fixed point can exist uniquely without all iterates converging to it.

3. Epigraph: $\min t$ subject to $x + 1 \le t$, $-2x + 4 \le t$, with x, t free. Dual: $\max p_1 + 4p_2$ subject to $p_1 - 2p_2 = 0$, $p_1 + p_2 = 1$, $p \ge 0$. The intersection gives $x^* = 1, t^* = 2$; $p = (2/3, 1/3)$ gives the matching value 2.

4. No. The active slopes are $1, 2$, whose convex hull $[1, 2]$ excludes zero. In fact, for $x < 0$, $f(x) = x$, so the problem is unbounded below. Ties alone do not prove optimality.

5. If $0 = \sum_i p_i a_i$ with $p \in \Delta_m$, then $f(x) \ge -\sum_i p_i b_i$ for every x, and $f(0)$ is finite. Conversely, if the compact convex hull excludes 0, strict separation gives a direction d and $\epsilon > 0$ with $a_i^\mathsf{T} d \le -\epsilon$ for every i. Then $f(td) \le -t\epsilon + \max_i(-b_i) \to -\infty$.

6. $x^* = (3, 2)$, with cost $2 \cdot 3 + 3 \cdot 2 = 12$. Symmetry makes $y = x^*$ dual feasible, and its value is also 12. The matrix is indefinite; that is irrelevant to this LP certificate.

7. (i) $Ax \ge (1, 1)^\mathsf{T}$; each coordinate can reach 1. (ii) $Ax \le (0, 0)^\mathsf{T}$; $b = 0$ is allowed and is the smallest RHS componentwise. A $\ge$ constraint fears a larger RHS, while a $\le$ constraint fears a smaller one.

8. The counterpart is $x_1 \le 5$, $x_2 \le 5$. The scalar dual satisfies $y \ge x_1$, $y \ge x_2$, $y \le 5$; such y exists exactly under those two inequalities.

9. The support is $\max(0, -x_1, -x_2)$. Thus $\beta \ge 0$ and $x_1, x_2 \ge -\beta$. For $\beta = -1$, no x works because $a = 0$ is allowed and would require $0 \le -1$.

10. Precisely $x_1 \le 3$ and $x_2 \le 0$. If $x_2 > 0$, the adversary can send a_2 to infinity. If the set is empty, the universal constraint is vacuous and imposes no restriction on x.

11. $a_0^\mathsf{T} x + 2\|x\|_2 \le 7$. Cauchy–Schwarz bounds $u^\mathsf{T} x \le 2\|x\|_2$, with equality at $u = 2x/\|x\|_2$ for $x \ne 0$; $x = 0$ is immediate.

Key idea. When a proof feels abstract, ask what the certificate is and how to check it: an invariant vector, a balancing probability mixture, a matching dual solution, or a finite set of multipliers bounding every uncertainty scenario.

12 One-page reference sheet

Duality. Feasible primal + feasible dual + equal objectives ⇒ both optimal. A feasible LP with finite value has an attained optimum and an equal attained dual optimum. Dual infeasibility requires a primal-feasibility check before concluding unboundedness.

Stochastic matrices. $P \geq 0$, $P\mathbf{1} = \mathbf{1}$. A probability column evolves as $\pi_{k+1} = P^{\mathsf{T}}\pi_k$. Stationarity: $P^{\mathsf{T}}\pi = \pi$, $\pi \geq 0$, $\mathbf{1}^{\mathsf{T}}\pi = 1$.

$$\max_{\lambda \geq 0}\{\mathbf{1}^{\mathsf{T}}\lambda : (P^{\mathsf{T}} - I)\lambda = 0\} \longleftrightarrow \min_{y}\{0 : (P - I)y \geq \mathbf{1}\}.$$

The dual fails at a largest coordinate of y, forcing a nonzero primal vector. Normalize it. Existence $\neq$ uniqueness $\neq$ convergence.

Maximum of affine functions.

$$\min_{x}\max_{i}(a_i^{\mathsf{T}}x - b_i) = \min_{x,t}\{t : Ax - t\mathbf{1} \leq b\} = \max_{p \geq 0}\{-b^{\mathsf{T}}p : A^{\mathsf{T}}p = 0,\ \mathbf{1}^{\mathsf{T}}p = 1\},$$

with finite value assumed and x, t free. Lower bound: $\max_i(a_i^{\mathsf{T}}x - b_i) \geq \sum_i p_i(a_i^{\mathsf{T}}x - b_i) = -b^{\mathsf{T}}p$. Optimal weights live on active pieces. Optimality at x: $0 \in \operatorname{conv}\{a_i : i \in I(x)\} = \partial f(x)$. Finite value iff $0 \in \operatorname{conv}\{a_1, \ldots, a_m\}$.

Symmetry certificate. For $\min c^{\mathsf{T}}x$ s.t. $Ax \geq c$, $x \geq 0$, if $A = A^{\mathsf{T}}$, $Ax^* = c$, $x^* \geq 0$, use dual $y = x^*$. Directly: $c^{\mathsf{T}}x = (x^*)^{\mathsf{T}}Ax \geq (x^*)^{\mathsf{T}}c$.

Robust RHS. For nonempty $\mathcal{U}$:

$$Ax \geq b\ \forall b \in \mathcal{U} \iff (Ax)_i \geq \sup_{b \in \mathcal{U}} b_i; \quad Ax \leq b\ \forall b \in \mathcal{U} \iff (Ax)_i \leq \inf_{b \in \mathcal{U}} b_i.$$

The extrema are coordinatewise; the assembled vector need not lie in $\mathcal{U}$. For $\mathcal{U} = \{b \geq 0 : \mathbf{1}^{\mathsf{T}}b \leq 1\}$, the first counterpart is $Ax \geq \mathbf{1}$.

Robust coefficients. Replace $a^{\mathsf{T}}x \leq \beta$ for all $a \in \mathcal{U}$ by $\sigma_{\mathcal{U}}(x) \leq \beta$. For the nonempty sets below, exact finite certificates are:

$$\mathcal{U} = \{a : Da = d,\ a \geq 0\}: \quad \exists y \text{ free}: \ D^{\mathsf{T}}y \geq x,\ d^{\mathsf{T}}y \leq \beta;$$
$$\mathcal{U} = \{a : Da \geq d,\ a \leq 0\}: \quad \exists \mu \geq 0: \ D^{\mathsf{T}}\mu \geq -x,\ -d^{\mathsf{T}}\mu \leq \beta.$$

Empty uncertainty ⇒ omit its block. Unbounded support value ⇒ that decision fails robustness. Unbounded set alone is not enough to conclude failure.

Final checks. Dimensions; free versus signed variables; min versus max; $\leq$ versus $\geq$; nonempty uncertainty; exactness via strong duality; candidate feasibility; matching objective values.

Companions: `hw04_solutions.pdf` contains full worked proofs.
`verify.py` checks examples; `verification_results.json` records the results.